From a practical perspective, the adjustment for the DO seems with 6% relatively small, especially in comparison with the adjustment of the FX component discussed next. Hence, in the absence of a better alternative, we usually assume this value of 6% – while seeing in every sovereign default the positive side of an opportunity to improve this estimate further.

Difference in Settlement Currency

In order to illustrate the value of the FX component, imagine we are long a five-year (5Y) asset&basis swapped JGB and a CDS on Japan, and after three years into the trade Japan defaults (Figure 13.1). We can deliver the JGB into the CDS and receive our initially invested USD principal back. However, we still need to serve all cash flows in the outstanding swap agreements. In particular, at maturity of the cross-currency basis swap (CCBS), we need to pay the JPY principal, which we thought to match with the JGB principal repayment. Since we now lost our JGB (to receive USD ahead of time), we need to replicate the JPY cash flows by exchanging the USD we received from the CDS back into JPY and invest that capital again in JPY in order to generate the JPY cash flows needed to serve the remaining payments in the outstanding ASW and BSW. Thus, entering into an asset&basis swapped JGB

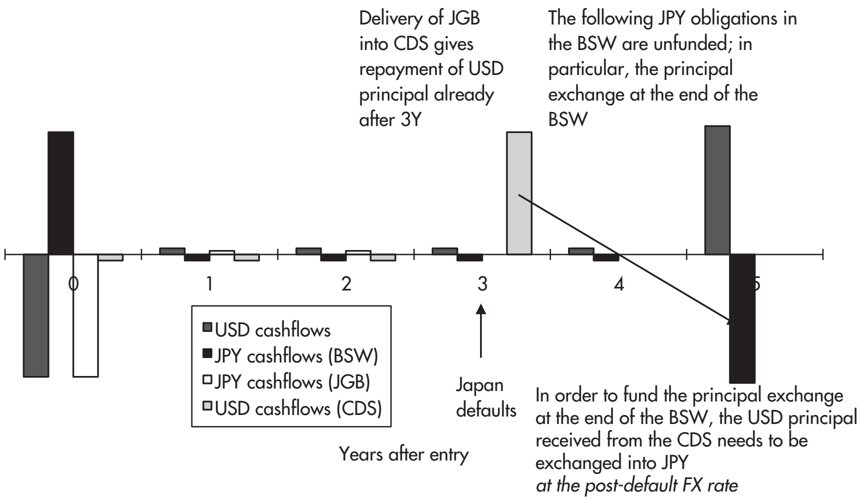

FIGURE 13.1 Cash flows in an investment into a 5Y asset&basis swapped and default protected JGB, with default after three years.
Source: Authors.

and a CDS on Japan includes entering into an *exposure to a JPY weakening conditional on a Japan default*. If the JPY weakens, say, from 80 to 160, following default, we can reproduce the JPY cash flows[4] we need to serve the CCBS with just half the USD we received from the CDS payment, keeping the remaining half as extra profit. This also affects the arbitrage positions between asset&basis swapped bonds and the CDS discussed in Chapter 16.

Similarly to the DO, when pricing the portion of the CDS due to currency effects, we face two issues:

- which pricing model to use;
- which parameters to use (e.g. which impact of credit risk on the FX rate should we assume?).

While we do not intend to participate in the ongoing academic discussion about the pricing model, we propose an original approach, modeled on the delta hedging of an option. The goal is not to present a solution but rather to gain a framework for illustrating the problems.

As starting point for the construction, we buy a USD-denominated CDS on Japan and sell a JPY-denominated CDS on Japan with the same end dates. Then, in case of a default, we can take the bond we are delivered and deliver it against receiving USD at a pre-defined FX rate. Assuming the yen weakens in the event of a default, we realize the difference between the pre- and post-default FX rate as profit. For example, if the JPY was to weaken from a pre-defined 80 to 160 in the event of default, our CDS spread position would be a binary option which pays us $50 in the event Japan defaults. Then, the fair difference between the CDS settling in USD and the CDS settling in JPY would be given as the annuity value of the fair value of that binary option.

Alternatively to pricing a binary option, we could replicate the delta hedging of the Black–Scholes framework, increasing a long JPY/short USD position as the credit risk of Japan (as measured by the CDS) increases. Then, in the event of no default, we have no FX position. But in the event of default (i.e. a default probability of one), our FX hedge covers the full long USD/short JPY position we get from settling the CDS spread. As in the case of delta hedging an option, this transfers the profit from the payoff at expiry to the profit from continuous adjustments of the hedge ratio. We can buy JPY when the default risk of Japan is high (and thus the JPY is expected to be cheap) and sell

[4] Of course, we face the risk that we cannot generate JPY cash flows at the original yield level. However, the gain from a weakening currency in case of its issuing government defaulting is likely to be much bigger than any potential losses from moves on the yield side.

JPY when the default risk of Japan is low (and thus the JPY is expected to be rich). Consequently, the fair value for the CDS spread (i.e. for the overcompensation through USD-denominated CDS) is a function of the *default probability volatility*. In other words, the higher the volatility of the CDS, the higher the premium of USD-denominated versus JPY-denominated CDS should be.[5]

Given the current stage of modeling the CDS, to our knowledge, there remain significant hurdles on the way to a consistent pricing of the overcompensation through USD as settlement currency (which we also refer to as the "FX component"). In particular, as our delta-hedging approach does not seem to have entered the academic discussion, it remains unclear whether and how it can be reconciled with the binary option approach.

An even bigger problem is posed by the estimation of the input variables in any model, just as in the case of the DO (where at least the model issue could be adequately addressed):

- How much should the currency of a defaulting country be expected to weaken?
- Does this weakening occur linearly as the credit risk increases, or are there jumps in the process?
- How can influences on the FX rate other than credit risk be excluded from the analysis?[6]
- What is the correlation between the default probability (and thus the FX rate) and the volatility of the default probability?

Again, one could look at historical precedents of sovereign defaults. But it is of no practical value to use the weakening of the Argentine peso as an estimate for the weakening of the JPY in case of a default, given the difference in foreign reserves, among others. Actually, if someone were to argue for the JPY to appreciate against the USD in case of Japan defaulting, due to Japan being forced to repatriate its large USD reserves, we could not refute this argument with certainty.

If it is possible to reasonably doubt even the *direction* of the currency movements of defaulting developed countries, there is little hope of achieving consensus about the *size* of the weakening of a currency in the event of default. Actually, one might even try the other way round. If quotes for

[5] As far as we are aware, this approach has not been discussed or formalized. Its implications for CDS pricing therefore remain unknown. However, our preliminary model based on this approach suggests that it could be a major driving force for CDS markets which exhibit a high volatility, such as Portugal during the euro crisis.

[6] Imagine that in the CDS spread construction from above the Japanese credit risk does not change but that the USD weakens due to an unrelated factor (e.g., China selling its US Treasuries).

the JPY-denominated CDS were available, one could translate the spread between the USD-denominated and the JPY-denominated CDS on Japan into the market-implied FX level in the event of default.[7]

Altogether, for pricing the FX component there is neither a consensus about the model nor an estimate for its parameters. This may explain the low liquidity for CDS without FX component, for example, for CDS on Japan settling in JPY. The scarce observations for these CDS without FX component known to us fall in a range between 30% and 60% of the quotes for CDS with FX component, i.e. the standard specifications. For example, if the usual CDS on Japan settling in USD trades at 100 bp, the CDS settling in JPY may be quoted in a range between 30 bp and 60 bp – which is a rare event and leads to two conclusions:

- Using the middle of the range, its level of only 45% suggests that the FX component is indeed a major factor in sovereign CDS prices. Adjusting for it – however imperfectly – is therefore vital. If we were simply using the unadjusted CDS quotes in our ASW model, we are likely to overestimate the actual credit risk of bonds by more than twice the actual amount. In the example of Japan above, our model would overestimate the fair JGB ASW by 55 bp, i.e. the amount of the CDS on Japan which reflects the FX component rather than the credit exposure of an asset swapped JGB, which is only 45 bp (ignoring the DO for the moment).
- The large width of the range may be a consequence of the missing pricing model, making traders reluctant to quote CDS without FX component at all; and in case they are forced to make an exception, they have little guidance from a pricing tool, need to apply a rough rule of thumb, with the results being widely distributed between 30% and 60%. Hence, using the middle of the range to adjust the CDS quotes in our ASW model is very likely to be an imprecise estimate. In case of an evolution of the pricing models allowing traders to quote CDS without FX component more frequently and more precisely, this will be a welcome opportunity to improve the adjustment.

Extracting the 'Pure' Credit Information from CDS Quotes by Adjusting for the DO and FX Component

The advantage of the CDS (versus fundamental credit analysis) is that it is a *quantitative*, traded variable. The disadvantage is that it is a variable that combines credit information with other elements. Before using the

[7]Under some assumptions for the probability distribution of default.

CDS quotes as input variable in RV relationships such as our ASW model, it is therefore necessary to extract the pure credit information from the CDS by adjusting for the fair value of its other elements, the DO and the overcompensation from USD as settlement currency. We found that for both adjustments – though they are in principle observable ex ante, unlike the variable cost of equity – there are only very few observations. Still, in the absence of a better alternative (such as a fundamental model for the FX component), we need to base our estimations on them and aim for gradual improvements as more and more observations[8] become available.

Given the information currently available to us, we might use the following adjustment to the CDS prices (with FX component) from the market in order to translate observable market prices in an estimate for the 'pure' credit risk of sovereign issuers, i.e. the input variable required in our swap spread model:

- All sovereign CDS with an FX component, i.e. all CDS usually traded, should be adjusted for this element in their prices, which is estimated to amount on average to 55% of the CDS quote observed. Hence, the CDS adjusted for the FX component = CDS quote observed (with FX component) * 0.45.
- Sovereign CDS with physical delivery, i.e. most of them, also need to be adjusted for the DO, which is estimated to amount to 6% of the CDS quote observed. Assuming independence between the DO and the FX component, another 6% of the full CDS quote need to be subtracted from the CDS adjusted for the FX component. Hence, the CDS adjusted for the FX component and DO = CDS quote observed (with FX component and DO) * 0.39.

Some RV relationships such as the ASW model require a quantitative estimation of the sovereign credit risk. Given the uncertainty of the quantification in an environment of scarce observations, it seems prudent to assess hedges and trades based on these RV relationships under several estimations for the DO and FX component. When thinking about a JGB swap spread trade in the example above, one could price it under three different input variables for the 'adjusted CDS,' at the bottom (30 bp), in the middle (45 bp), and at the top (60 bp) of the range. If it appeared attractive under all three scenarios, we would feel more comfortable than if it only worked at the bottom of the range.

[8]It is possible that analysts with good access to CDS traders can already produce better estimates from more observations than the authors.

Other RV relationships rely on qualitative statements only. For example, even without quantifying the DO or the FX component, we can say that neither of the two should be negative and that thus the 'pure' credit risk should not be higher than the CDS quote representing the combination of credit risk, DO, and FX component. The arbitrage inequality presented in Chapter 16 will only rely on such a qualitative statement and thereby remain unaffected by the quantification problems, which will contribute to its firmness.

OTHER APPLICATIONS OF CDS: TRADING CDS VERSUS OTHER CDS AND VERSUS BONDS

Apart from fulfilling the function as input variable for credit exposure in pricing formulae, CDS are interesting both as a separate class of financial instruments and in relation to their underlying bonds. These two approaches are different from a conceptual perspective, lead to different trades and are covered in different parts of the book: the rest of this chapter looks at the intrinsic relationships of CDS as separate asset class, while Chapter 16 will connect sovereign CDS with the sovereign bonds asset&basis swapped into USD.

Conceptually, one can price any instrument either versus instruments of the same type or versus other instruments. In the first case, one evaluates the intrinsic, typically statistical, properties of the instrument; in the latter, one assesses how the same information is expressed through different instruments. We can use this conceptual difference to classify CDS pricing models and trades (and shall use it later on to classify option pricing models and trades):

- In this chapter, we analyze the CDS in isolation of its relationship to other markets. Thus, we are mainly concerned with the statistical properties between CDS quotes, for example, with the factor structure of a CDS curve (different maturities of a CDS on the same issuer) and with the factor structure of the CDS market on euro sovereign issuers (same maturity of CDS on different euro governments). As far as we are aware, this approach is less commonly applied by market participants. Consequently, we often find more relative value opportunities in these 'neglected' statistical relationships within the CDS market than in the well-analyzed link of the CDS to swap spreads.
- Since both bond yields and CDS quotes are driven to some extent by the credit quality of the bond issuer, one can compare the way the information is expressed in both markets, derive the 'fair' CDS quote by the bond yields (or, vice versa, the 'fair' bond yields by the CDS quotes), and exploit

potential mismatches by trading bonds against CDS. Because CDS express the sovereign default risk as a spread over USD SOFR, this comparison needs to be done against the SOFR-ASW of sovereign issues asset&basis swapped into USD. Turning this general concept into a trading strategy that works in practice requires taking into account the elements other than credit that also impact swap spreads and CDS. This work will be completed in Chapter 16, after the basis swaps will have been introduced in the next chapters.

Before starting with the statistical analysis of CDS, a word of caution is required: when considering CDS in isolation from its relationship to bonds, it is easy to lose sight of the issues involved in the event of default. And in fact, sometimes the CDS market seems to trade *as if these issues did not exist*. For example, trading occurs without the two partners being concerned about the inability to price the FX component. And as long as there is no default, this abstract treatment of CDS as a statistical time series without connection to bonds (thus also in abstraction from the problems of the connection between CDS and bonds) works well. A 5Y-10Y CDS curve trade, for example, can be assessed by its statistical properties and traded just like a yield curve position. And if mean reversion occurs before default, it is unlikely that this abstraction will cause any problems. However, in the event of a default, the abstract treatment of CDS can no longer be maintained; when faced with the physical delivery of a bond, at the latest, the relationship of a CDS to the bond market can no longer be ignored.

Accordingly, we shall begin our treatment of CDS on an abstract basis, using statistical methods to derive trading strategies within the CDS market. Then, we shall analyze the way those strategies are likely to fare in the event of a default and restrict our positions to those that can be expected to perform well in the event of no default (due to their statistical properties) and to be unaffected by potential problems in case of a default (e.g. by not being short the DO).

In addition, every comprehensive CDS analysis should include fundamental credit analysis. While this is outside the scope of this book, it should be part of the assessment of credit risk.

A PCA ON THE CDS CURVE

Figure 13.2, Figure 13.3, and Figure 13.4 depict the results of a PCA of the Italian CDS curve during the euro crisis (2Y, 5Y, and 10Y USD-denominated CDS quotes, weekly level data from January 2006 to August 2012).

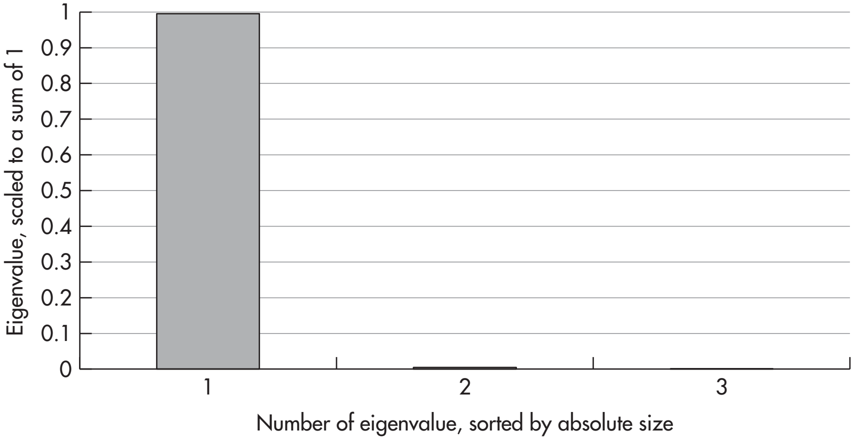

FIGURE 13.2 Scaled eigenvalues of a PCA on the Italian CDS curve.
Sources: data – Authors, Bloomberg; chart – Authors.

Data period: 1 Jan 2006 to 26 Aug 2012, weekly data.

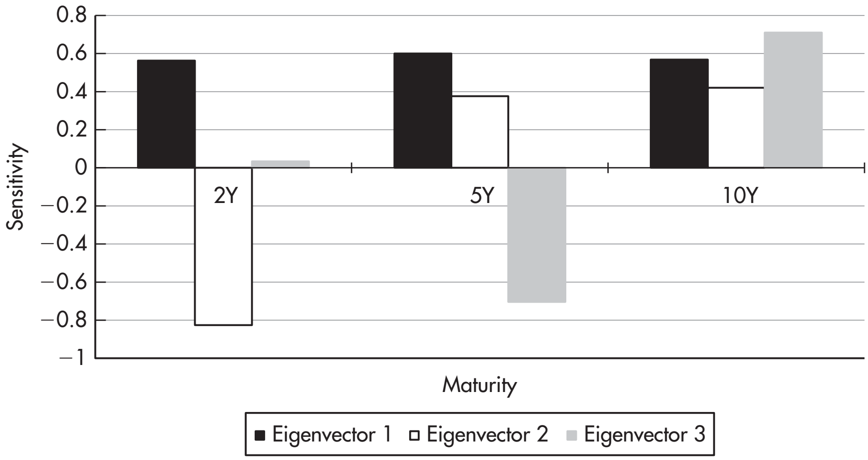

FIGURE 13.3 First three eigenvectors of a PCA on the Italian CDS curve.
Sources: data – Authors, Bloomberg; chart – Authors.

Data period: 1 Jan 2006 to 26 Aug 2012, weekly data.

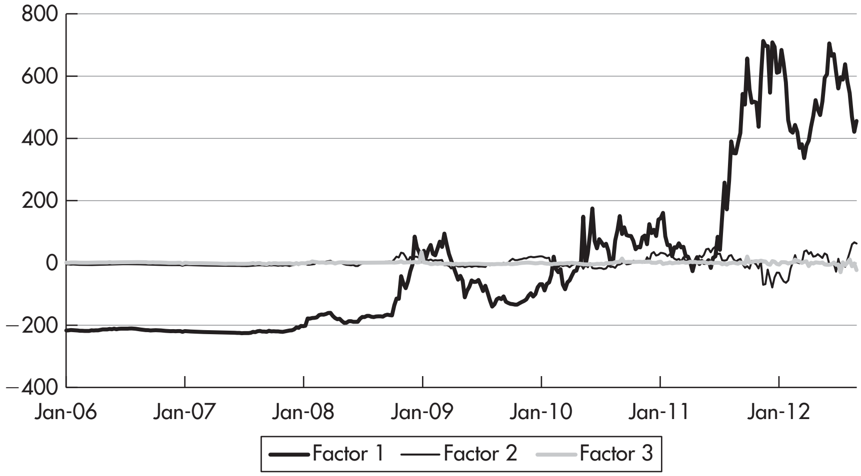

FIGURE 13.4 First three factors of a PCA on the Italian CDS curve.
Sources: data – Authors, Bloomberg; chart – Authors.

Data period: 1 Jan 2006 to 26 Aug 2012, weekly data.

It turns out that the first factor explains 99.5% of the overall variation across the CDS curve. This is a typical result for a PCA run on CDS curves (same issuer, different maturities) and indicates that CDS curves exhibit basically a single-factor structure. It means that the whole CDS curve contains in essence just one piece of information (factor 1, the overall CDS level), and that, given one maturity on the CDS curve, the whole maturity spectrum can be reconstructed with high accuracy (by the sensitivities of the first eigenvector). This statistical property is in line with anecdotal evidence about the pricing of CDS traders. They tend to focus on the most liquid maturity (usually 5Y) and to adjust their quotes for other maturities as a linear function of the moves in the most liquid one.

Like the example of Italy above, CDS curves typically display a single-factor structure, which is indicative of a market in an early stage of development.[9] The theoretical conclusion is that the credit risk element in bond yields cannot explain the three-factor structure observed in bond yield curves.

Thus, the three-factor structure of bond yield curves must arise from the risk-free yield curve (e.g. from inflation expectations) rather than from the default risk element.

[9] See Chapter 3 for more details on PCA.

The practical consequence for trading the CDS curve is that there is little scope for relative value trades across CDS curves. For example, a 5Y-10Y CDS curve trade *unaffected* by directional impacts on the CDS curve (i.e. a position on factor 2) faces the problem of a low range of factor 2 (since 99.5% of the variance across the CDS curve is already explained by directional impacts), which means that the potential profit from relative value trades on the CDS curve is usually too small to cover the relatively high bid–offer spreads. Hence, relative value considerations can only play a role in selecting the best maturities for expressing directional views on the CDS curve. In the example above, an investor who wants to position for a decrease of the Italian CDS in general (factor 1 decreasing) can enhance his profit by expressing his view through selling 5Y or 10Y rather than 2Y Italian CDS (given the shape of the second eigenvector and the fact that factor 2 is positive).

While a single-factor structure for the CDS curve as in the example above is typical, there are some exceptions when more than just one point on the CDS curve is actively and independently traded. In these instances, one can observe a richer factor structure and deploy the entire arsenal of statistical yield curve analysis outlined in Chapter 2 and Chapter 3 to the CDS curve as well.

Among relative value traders focusing on the CDS, it is common to strip out the implied default probabilities from CDS curves and scan the result for anomalies, in particular for negative implied default probabilities. While there is nothing wrong with this approach, we recommend complementing it with the additional perspective from a statistical analysis along the lines above.

A PCA ON THE EUR SOVEREIGN CDS UNIVERSE

While CDS curves on different maturities of the same issuer usually display single-factor structures, a set of CDS of different issuers, with the same maturities, can exhibit a richer factor structure and thus allow a wider variety of trades, in particular also relative value trades, which are hedged against factor 1 (i.e. the overall level of CDS quotes) and exploit meaningful higher factors. A natural application for a PCA on CDS of the same maturity (here 5Y) covering different sovereign issuers is an analysis of the euro universe (again using the time period of the euro crisis as a valuable case study), with the results being shown in Figure 13.5 and Figure 13.6.

It turns out that the first factor, explaining 92% of overall variation in the EUR sovereign CDS market, represents the overall level of CDS in the Eurozone. Thus, factor 1 can serve as a measure of the market's perception of sovereign credit risk in the Eurozone in general. The sensitivities to factor 1, given by the first eigenvector, show the extent to which a specific country is affected by a general worsening (or improvement) of the euro crisis.